

# ETH Z - Disk Mat HS 24

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## ② Propositional Logic

D2.1 mathematical statement (= proposition) is true or false

Theorem/Lemma/Corollary: math. statements, known to be true, Conjecture: not known (but believed) to be true or false, assumption: not known but believed to be true

D2.3 logical values (constants): true (1) / false (0)

D2.4 negation (NOT):  $\neg$ , conjunction (AND):  $\wedge$ , disjunction (OR):  $\vee$

D2.5 formula: correctly formed expression with propositional symbols  $A, B, C$  and logical operators

implication:  $A \rightarrow B \equiv \neg A \vee B$

two-sided implication:  $A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$

D2.6 equivalent:  $F \equiv G$ , truth values are equal for all truth assignments

L2.1 same as L6.1, 1-8

D2.7 logical consequence:  $F \models G$ ,  $G$  is 1 if  $F$  is 1

D2.8 tautology (= valid): formula true for all truth assignments,  $\models F$

D2.9 satisfiable: true for at least 1 truth assignment, unsatisfiable otherwise

L2.2  $F$  is tautology iff  $\neg F$  is unsatisfiable

L2.3  $F \rightarrow G$  is tautology iff  $F \models G$

## ② Predicate Logic

D2.10  $k$ -ary predicate  $P$  on  $U$  is a function  $U^k \rightarrow \{0,1\}$

D2.11  $\forall x F$  is true iff  $F$  is true for all  $x$  in  $U$

$\exists x F$  is true iff  $F$  is true for some  $x$  in  $U$

prime( $x$ )  $\Leftrightarrow x > 1 \wedge \forall y \forall z ((y \cdot z = x) \Rightarrow (y=1) \vee (z=1))$

L2.4 if  $F \models G$ , then " $F$  is valid  $\Rightarrow G$  is valid" is true

## ② Proof Patterns

D2.12 Composition of implications if  $S \Rightarrow T$  is true

and  $T \Rightarrow U$  is true,  $S \Rightarrow U$  is also true

D2.13 Direct proof of an implication  $S \Rightarrow T$ , assume  $S$ , prove  $T$  under the assumption

D2.14 Indirect proof of an implication  $S \Rightarrow T$ , assume  $T$  is false and prove  $S$  is false under the assumption

D2.15 Modus Ponens ( $S$ ): Find  $R$ , prove  $R$ , prove  $R \Rightarrow S$

D2.16 Case Distinction ( $S$ ): "Split"  $S$  into finite list  $R_1, \dots, R_n$  of cases, prove at least one occurs (mostly trivial), prove  $R_i \Rightarrow S$

D2.17 Proof by contradiction ( $S$ ): Find  $T$ , prove  $T$  is false assume  $S$  is false and prove  $T$  is true under the assumption (often: Assume  $S$  is false, derive trivially wrong  $T$ )

D2.18 Existence Proof: proof  $S_x$  is true for at least one  $x \in \mathcal{X}$ , constructive: show an example, non-constructive otherwise

T2.10 Pigeonhole Principle:  $n$  objects partitioned into  $k < n$  sets, at least one set contains  $\lceil \frac{n}{k} \rceil$  objects (existence proof)

D2.19 Proof by Counterexample: proof  $S_x$  is not true for all  $x \in \mathcal{X}$

Proof by induction: 1. Basis step: proof  $P(0)$ , 2. Induction step: proof for arbitrary  $n$   $P(n) \Rightarrow P(n+1)$

T2.11 for  $U = \mathbb{N}$  and any unary  $P$  we have

$P(0) \wedge \forall n (P(n) \Rightarrow P(n+1)) \Rightarrow \forall n P(n)$

## ③ Sets

D3.1 (informal) cardinality: number of elements in finite set, denoted  $|A|$

Russell's Paradox: Set  $R = \{A \mid A \in A\}$

Either  $R \in R$  or  $R \notin R \rightarrow$  paradox, axiomatic set theory (ZF):  $U$  not a set, never  $A \in A$

elementhood predicate:  $\text{inf } x \in y, \notin$  for  $\neg(x \in y)$

D3.2 equality:  $A = B \Leftrightarrow \forall x (x \in A \Leftrightarrow x \in B)$

L3.1 For any  $a$  and  $b$ ,  $\{a\} = \{b\} \Rightarrow a = b$

Elements in a set have no order:  $\{a, b\} = \{b, a\}$

Ordered pairs:  $(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$

$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}$  (technicality)

D3.3 Subset  $A \subseteq B \Leftrightarrow \forall x (x \in A \Rightarrow x \in B)$

L3.2 equality  $A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

L3.3 Subset relation is transitive  $A \subseteq B, B \subseteq C \Rightarrow A \subseteq C$

D3.4 union:  $A \cup B \stackrel{\text{def}}{=} \{x \mid x \in A \vee x \in B\}$

intersection:  $A \cap B \stackrel{\text{def}}{=} \{x \mid x \in A \wedge x \in B\}$

Let  $A$  be a non-empty set of sets:

$\bigcup A \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for some } A \in A\}$

$\bigcap A \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for all } A \in A\}$

D3.5 difference  $B \setminus A \stackrel{\text{def}}{=} \{x \in B \mid x \notin A\}$

T3.4 Idempotence:  $A \cap A = A, A \cup A = A$

Commutativity:  $A \cap B = B \cap A, A \cup B = B \cup A$

Associativity:  $A \cap (B \cap C) = (A \cap B) \cap C,$

$A \cup (B \cup C) = (A \cup B) \cup C$

Absorption:  $A \cap (A \cup B) = A, A \cup (A \cap B) = A$

Distributivity:  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Consistency:  $A \subseteq B \Leftrightarrow A \cap B = A \Leftrightarrow A \cup B = B$

D3.6 Empty set: contains no element:  $\forall x \neg(x \in \emptyset)$

L3.5 There is only one empty set denoted  $\emptyset$  or  $\{\}$

L3.6 The Empty set is subset of every set:  $\forall A (\emptyset \subseteq A)$

One could construct a definition of the natural numbers as follows:  $0 \stackrel{\text{def}}{=} \emptyset, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}$

D3.7 Power Set:  $\mathcal{P}(A) \stackrel{\text{def}}{=} \{S \mid S \subseteq A\}$

For finite set, power set has cardinality  $2^k$ , therefore an alternative notation is  $2^A (= \mathcal{P}(A))$

D3.8 Cartesian product  $A \times B = \{(a, b) \mid a \in A, b \in B\}$

For finite sets:  $|A \times B| = |A| \cdot |B|$

Generally for  $k$ -tuples  $X_{i=1}^k A_i = \{(a_1, \dots, a_k) \mid a_i \in A_i \text{ for } 1 \leq i \leq k\}$

The cartesian product is not associative:  $X_{i=1}^3 A_i \neq (A_1 \times A_2) \times A_3$

## ③ Relations

D3.9 A (binary) relation  $\rho$  from set  $A$  to set  $B$  is a subset of  $A \times B$ , if  $A = B$ , then  $\rho$  is a relation on  $A$

Instead of  $(a, b) \in \rho$  one usually writes  $a \rho b$  (resp.  $a \not\rho b$ )

Relation  $\equiv_m$  on  $\mathbb{Z}$ :  $a \equiv_m b \Leftrightarrow a - b = km$  for some  $k$

Empty relation:  $\emptyset$ , complete relation:  $A \times B$

D3.10 identity relation on  $A$ :  $\text{id}_A = \{(a, a) \mid a \in A\}$

There are  $2^{n^2}$  different relations on  $A$ , if  $|A| = n$

One can represent a (binary) relation  $\rho$  as a

Boolean  $|A| \times |B|$  matrix  $M^\rho$ , 1 if  $a \rho b$ , 0 otherwise  
 $A = \{a, b\}, B = \{c, d\}, \rho = \{(a, c), (b, d), (a, d)\}$

$M^\rho = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$   $a \rightarrow c$   
 $b \rightarrow d$

An alternative representation is a directed graph

D3.11 inverse of  $\rho$ :  $\rho^{-1} = \{(b, a) \mid (a, b) \in \rho\}$  (alternative notation  $\rho^{-1}$ )

D3.12 composition of  $\rho$  and  $\sigma$ :

$\rho \circ \sigma \stackrel{\text{def}}{=} \{(a, c) \mid \exists b ((a, b) \in \rho \wedge (b, c) \in \sigma)\}$

$n$ -fold composition of  $\rho$  with itself is denoted  $\rho^n$

L3.7 composition is associative:  $\rho \circ (\sigma \circ \phi) = (\rho \circ \sigma) \circ \phi$

Short notation:  $\rho \circ \sigma$  instead of  $\rho \circ \sigma$

L3.8  $\rho \circ \sigma = \sigma \circ \rho$

D3.13  $\rho$  on  $A$  is reflexive if  $a \rho a$  for all  $a \in A$ , i.e.  $\text{id}_A \subseteq \rho$

D3.14  $\rho$  on  $A$  is irreflexive if  $a \not\rho a$  for all  $a \in A$ , i.e.  $\rho \cap \text{id}_A = \emptyset$

D3.15  $\rho$  on  $A$  is symmetric if  $a \rho b \Leftrightarrow b \rho a$ , i.e.  $\rho = \rho^{-1}$

D3.16  $\rho$  on  $A$  is antisymmetric if  $a \rho b \wedge b \rho a \Rightarrow a = b$ , i.e.  $\rho \cap \rho^{-1} = \text{id}_A$

D3.17  $\rho$  on  $A$  is transitive if  $a \rho b \wedge b \rho c \Rightarrow a \rho c$ , i.e. L3.9

L3.9  $\rho$  is transitive iff  $\rho^2 \subseteq \rho$

D3.18 transitive closure  $\rho^+ = \bigcup_{n \in \mathbb{N}} \rho^n$

D3.19 equivalence relation is reflexive, symmetric and transitive

D3.20 equivalence class: set of elements that are equivalent to  $a \in A$ :  $[a]_\theta \stackrel{\text{def}}{=} \{b \in A \mid b \theta a\}$  ( $\theta$  is an equivalence relation)

trivial equivalence relations are the complete relation  $A \times A$  with one equivalence class:  $A$ , and the identity relation for which the equivalence classes are all singletons  $\{a\}$  for all  $a \in A$

L3.10 The intersection of two equivalence relations is an equivalence relation

D3.21 partition: mutually disjoint subsets that cover  $A$  i.e.  $\{S_i \mid i \in I\}$  for some index set  $I$ ,  $S_i \cap S_j = \emptyset$  for  $i \neq j$  and  $\bigcup_{i \in I} S_i = A$

Partition and equivalence relation capture some abstraction

D3.22 quotient set of  $A$  by  $\theta$ : set of equivalence classes  $A/\theta \stackrel{\text{def}}{=} \{[a]_\theta \mid a \in A\}$  (also called  $A \text{ modulo } \theta$  or  $A \text{ mod } \theta$ )

T3.11  $A/\theta$  is a partition of  $A$

One could construct a definition of the rational numbers  $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ , equivalence relation  $\sim$  on  $A$

$(a, b) \sim (c, d) \Leftrightarrow ad = bc$ ,  $\mathbb{Q} \stackrel{\text{def}}{=} (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})) / \sim$

### ③ Partial Order Relations

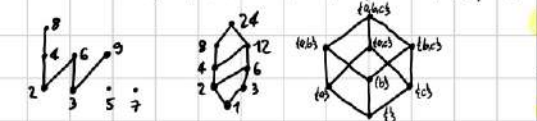
D323 A partial order (or order relation) is reflexive, antisymmetric and transitive, A together with partial order  $\leq$  is a partially ordered set (short: poset), denoted  $(A; \leq)$   
 $a < b \stackrel{\text{def}}{\iff} a \leq b \wedge a \neq b$

D324 a and b are comparable if  $a \leq b$  or  $b \leq a$ , otherwise: incomparable

D325 If any two elements are comparable, A is totally ordered (or linearly ordered) by  $\leq$

D326 b covers a if  $a < b$  and there exists no c with  $a < c$  and  $c < b$  i.e. between a and b

D327 Hasse diagram: directed graph, with vertices for elements and edge  $a \rightarrow b$  iff b covers a  
 usually  $a < b \rightarrow b$  placed higher, arrows omitted  
 Examples:  $(\{2, 3, 4, 5, 6, 7, 8, 9\}; \mid)$ ,  $(\{1, 2, 3, 4, 6, 8, 12, 24\}; \mid)$ ,  $(\mathcal{P}(\{a, b, c\}); \subseteq)$



D328 direct product of posets  $(A; \leq) \times (B; \leq)$  is set  $A \times B$  with relation  $\leq$   
 $(a_1, b_1) \leq (a_2, b_2) \stackrel{\text{def}}{\iff} a_1 \leq a_2 \wedge b_1 \leq b_2$

T3.12  $(A; \leq) \times (B; \leq)$  is a poset

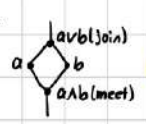
T3.13 lexicographic order  $\leq_{\text{lex}}$  defined on  $A \times B$ :  
 $(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \stackrel{\text{def}}{\iff} a_1 < a_2 \vee (a_1 = a_2 \wedge b_1 \leq b_2)$   
 If  $(A; \leq)$ ,  $(B; \leq)$  totally ordered,  $\leq_{\text{lex}}$  is as well  
 Can be generalized to k-tuples over some alphabet  $\Sigma$ , denoted  $\Sigma^*$  or generally  $\Sigma^*$  finite-length strings (example: telephone book)

D329 Let  $(A; \leq)$  be a poset,  $S \subseteq A$ ,  $a \in A$  is:

minimal (maximal) element if no  $b, b < a$  ( $b > a$ )  
 the least (greatest) element if  $a \leq b$  ( $a \geq b$ ) for all  $b$   
 lower (upper) bound of S if  $a \leq b$  ( $a \geq b$ ) for all  $b \in S$   
 greatest lower bound (least upper bound) of S if a is greatest (least) of the set of all lower (upper) bounds of S (sometimes denoted  $\text{glb}(S)$  /  $\text{lub}(S)$ )

D330 poset is well-ordered if totally ordered and every non-empty subset has a least element  
 Well-ordering is only of interest for infinite posets  
 Example:  $\mathbb{N}$  is well-ordered,  $\mathbb{Z}$  isn't

D331 If a and b have a greatest lower (least upper) bound, it's called meet (join), denoted  $a \wedge b$  ( $a \vee b$ )



D332 lattice: poset, every pair has meet and join

### ③ Functions

D333 function  $f: A \rightarrow B$  from domain A to codomain B is a relation from A to B  
 $\forall a \in A \exists b \in B a \{ b \}$  (f is totally defined)  
 $\forall a \in A \forall b, b' \in B (a \{ b \} \wedge a \{ b' \} \rightarrow b = b')$  (f is well defined)

usually denoted  $f(a)$ , to define  $f, f: a \mapsto \text{expr. in } a$

D334 set of all functions  $A \rightarrow B$  denoted  $B^A$   
 For finite sets there are  $|B|^{|A|}$  such functions

D335 partial function: relation, just well defined (not totally)  
 functions are equal if relations are equal

D336 image of  $S \subseteq A$  under  $f: S \mapsto \{f(a) \mid a \in S\}$

D337  $f(A) \subseteq B$  is called image (or range) of f, den.  $\text{Im}(f)$

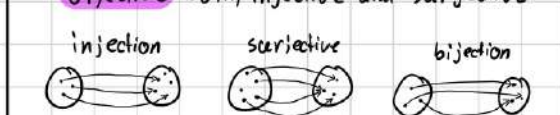
D338 preimage of  $T \subseteq B: f^{-1}(T) \stackrel{\text{def}}{=} \{a \in A \mid f(a) \in T\}$

D339 A function  $f: A \rightarrow B$  is:

injective (one-to-one):  $a \neq a' \rightarrow f(a) \neq f(a')$

surjective (onto): for every  $b \in B, b = f(a)$  for some  $a \in A$

bijective: both, injective and surjective



D340 For bijective function, inverse (same as relation) is called inverse function, denoted  $f^{-1}$

D341 composition of functions:  $f: A \rightarrow B, g: B \rightarrow C$   
 $g \circ f$  or  $gf$  defined by  $(g \circ f)(a) = g(f(a))$   
 not the same as for relations (ordering), declare it!

T3.14 Function composition is associative  $(h \circ g) \circ f = h \circ (g \circ f)$

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### ③ Countability of Sets

D342 i) Two sets are equinumerous (=gleich mächtig) denoted  $A \sim B$  if there exists a bijection  $A \rightarrow B$   
 ii) B dominates A denoted  $A \leq B$  if  $A \sim C$  for some  $C \subseteq B$  or equivalently injection  $A \rightarrow B$   
 iii) A is countable if  $A \leq \mathbb{N}$ , uncountable otherwise

T3.15 i)  $\sim$  is an equivalence relation  
 ii)  $\leq$  is transitive  
 iii)  $A \leq B \rightarrow A \leq B$

T3.16  $A \leq B \wedge B \leq A \Rightarrow A \sim B$

For finite sets  $A \sim B$  iff  $|A| = |B|$

T3.17 A is Countable iff it's finite or  $A \sim \mathbb{N}$

"There is no cardinality level between finite and countably infinite"

T3.18  $\{0, 1\}^* \stackrel{\text{def}}{=} \{\epsilon, 0, 1, 00, 01, \dots\}$  (finite) is countable

T3.19  $\mathbb{N} \times \mathbb{N} (= \mathbb{N}^2)$  is countable

C3.20  $A \times B$  of two countable sets is countable

C3.21  $\mathbb{Q}$  is countable

T3.22 Let  $A_i$  for  $i \in \mathbb{N}$  be countable:

i)  $A^n, n \in \mathbb{N}$  is countable

ii)  $\bigcup_{i \in \mathbb{N}} A_i$  is countable

iii)  $A^*$ , a finite sequence of elements A is countable

D3.43  $\{0, 1\}^\omega$ : semi-infinite binary sequences (or equivalently, the set of functions  $\mathbb{N} \rightarrow \{0, 1\}$ )

T3.23  $\{0, 1\}^\omega$  is uncountable

Proof: Cantor's diagonalization argument

Assume bijection  $f: \mathbb{N} \rightarrow \{0, 1\}^\omega$  exists

Let  $\beta_{ij}$  be the j-th bit in the i-th sequence  $f(i)$

$f(i) \stackrel{\text{def}}{=} \beta_{i0}, \beta_{i1}, \dots$

Let  $\bar{\beta}_i$  be the complement of the bit  $\beta_{ii}$

$\alpha \stackrel{\text{def}}{=} \bar{\beta}_{00}, \bar{\beta}_{11}, \dots$

$\alpha \in \{0, 1\}^\omega$ , there is no  $n \in \mathbb{N}$  such that  $\alpha = f(n)$ , since  $\alpha$  differs from every sequence by at least one bit  $\rightarrow f$  cannot be a bijection  $\square$

D3.44 A function  $f: \mathbb{N} \rightarrow \{0, 1\}$  is computable if there is a program, that with input n outputs  $f(n)$

C3.24 There are uncomputable functions  $\mathbb{N} \rightarrow \{0, 1\}$

Example: The halting problem is undecidable

### ④ Number Theory

Theory of  $\mathbb{N}$  or more generally  $\mathbb{Z}$

D4.1 a divides b:  $a \mid b$  if there exists a c so that  $b = ac$  ( $\Rightarrow c = \frac{b}{a}$ ), a: divisor of b, b: multiple of a, c: quotient (when b is divided by a),  $\nmid$ : not divide

every non-zero int is divisor of 0, 1 and -1 are divisors for every int

T4.1 For all a and  $d \neq 0$  exists a unique q and r:  $a = dq + r, 0 \leq r < |d|$

a: dividend, d: divisor, q: quotient, r: remainder

r is denoted  $R_d(a)$  or  $a \bmod d$

D4.2 Greatest common divisor d of a and b (not both 0)

$d \mid a \wedge d \mid b \wedge \forall c (c \mid a \wedge c \mid b \rightarrow c \mid d)$

For  $\mathbb{Z}$ : There are two:  $\pm d$

D4.3  $\text{gcd}(a, b)$  = unique positive greatest common divisor, if  $\text{gcd}(a, b) = 1$ , a and b are relatively prime

D4.2  $\text{gcd}(m, n - qm) = \text{gcd}(m, n)$

Euclid's gcd-algorithm:  $\text{gcd}(m, R_m(n)) = \text{gcd}(m, n)$

D4.4 ideal generated by a and b:  $(a, b) \stackrel{\text{def}}{=} \{ua + vb \mid u, v \in \mathbb{Z}\}$

similarly by single int a:  $(a) \stackrel{\text{def}}{=} \{ua \mid u \in \mathbb{Z}\}$

L4.3 For  $a, b \in \mathbb{Z}$  exists a  $d \in \mathbb{Z}$ , such that  $(a, b) = (d)$

L4.4 If  $(a, b) = (d)$ , d is a greatest common divisor of a and b

C4.5 For  $a, b \in \mathbb{Z}$  (not both 0) exist  $u, v \in \mathbb{Z}: \text{gcd}(a, b) = ua + vb$

D4.5 least common multiple L, denoted  $\text{lcm}(a, b)$ :

$a \mid L \wedge b \mid L \wedge \forall m (a \mid m \wedge b \mid m \rightarrow L \mid m)$

D4.6 prime:  $p > 1$  is prime if 1 and p are the only positive divisors of p, non-prime int  $> 1$ : composite

1 is neither prime nor composite

T4.6 Every positive int can uniquely be written as product of primes (1 has zero prime factors, which is allowed)

a and b can be written as:  $a = \prod p_i^{e_i}$  and  $b = \prod p_i^{f_i}$

$\text{gcd}(a, b) = \prod p_i^{\min(e_i, f_i)}, \text{lcm}(a, b) = \prod p_i^{\max(e_i, f_i)}$

$\text{gcd}(a, b) \cdot \text{lcm}(a, b) = a \cdot b$  (since  $\min(e_i, f_i) + \max(e_i, f_i) = e_i + f_i$ )

L4.12 Every composite integer n has a prime divisor  $\leq \sqrt{n}$

D4.8 modular congruence  $a \equiv_m b \stackrel{\text{def}}{\iff} m \mid (a - b)$

L4.3 For any  $m \geq 1, \equiv_m$  is an equivalence relation on  $\mathbb{Z}$

1.14 If  $a \equiv_m b$  and  $c \equiv_m d$ , then  $a+c \equiv_m b+d$  and  $ac \equiv_m bd$

1.15 Let  $f(x_1, \dots, x_k)$  be a multi-variate polynomial if  $a_i \equiv_m b_i$  for  $1 \leq i \leq k$ , then  $f(a_1, \dots, a_k) \equiv_m f(b_1, \dots, b_k)$

There are  $m$  equivalence classes of  $\equiv_m$ ,  $[0], [1], \dots, [m-1]$ , each class has representative  $R_m(a) \in [a]$  in the set  $\mathbb{Z}_m = \{0, \dots, m-1\}$

1.16 (i)  $a \equiv_m R_m(a)$ , (ii)  $a \equiv_m b \Leftrightarrow R_m(a) = R_m(b)$

Modular arithmetic: in computation with addition and multiplication, one can compute remainders at intermediate steps, instead of at the end

1.17 for multi-variate polynomial  $f$  and  $m \geq 1$ :

$$R_m(f(a_1, \dots, a_k)) = R_m(f(R_m(a_1), \dots, R_m(a_k)))$$

$R_m(n)$  can easily be computed by adding digits, then mod 9

1.18  $ax \equiv_m 1$  has a solution (unique) iff  $\gcd(a, m) = 1$

1.19 If  $\gcd(a, m) = 1$ , unique solution  $x \in \mathbb{Z}_m$  to  $ax \equiv_m 1$  is called multiplicative inverse of  $a$  modulo  $m$ , denoted  $x \equiv_m a^{-1}$  or  $x \equiv_m 1/a$

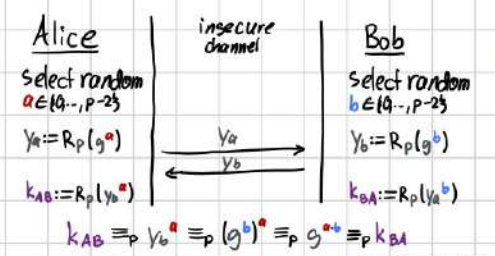
1.20 The Chinese Remainder Theorem:

Let  $m_1, m_2, \dots, m_r$  be relatively prime integers  $M = \prod_{i=1}^r m_i$ , For every list  $a_1, \dots, a_r$  with  $0 \leq a_i < m_i$ , the system  $x \equiv_{m_i} a_i$ ,  $1 \leq i \leq r$  for  $x$  has a unique solution  $0 \leq x < M$

To calculate  $x$ :  $M_i = M/m_i$ ,  $M_i N_i \equiv_m 1$   
 $x = R_m(\sum_{i=1}^r a_i M_i N_i)$ , Example: Compute  $R_{35}(2^{1000})$   
 $2^{1000} \equiv_5 1$  ( $2^4 \equiv_5 1$ ),  $2^{1000} \equiv_7 2$  ( $2^3 \equiv_7 1$ )  $\Rightarrow 2^{1000} \equiv_{35} 16$ , since 16 is the  $x \in [0, 34]$  with  $x \equiv_5 1$  and  $x \equiv_7 2$

Diffie-Hellman Key-Agreement:

public:  $p$ : (large) prime,  $g$ : basis (generator of  $\mathbb{Z}_p$ )



$x \mapsto R_p(g^x)$  is called one-way function: easy in one way, computationally infeasible

## 5 Algebra

D5.1 An operation on a set is a function  $S^n \rightarrow S$ , where  $n \geq 0$  is called "arity" (unary, binary...)  
 An operation with arity 0 is called constant

D5.2 An algebra (= algebraic structure, =  $\Omega$ -algebra) is a pair  $\langle S; \Omega \rangle$ ,  $S$  is a set (carrier),  $\Omega = \{\omega_1, \dots, \omega_n\}$  is a list of operations on  $S$

We sometimes drop unary and nullary operations if they are clear ( $\langle \mathbb{Z}; +; \cdot \rangle = \langle \mathbb{Z}; +; \cdot; 0; 1 \rangle$ )

D5.3 left (right) neutral element (or identity element):  $e \in S$ ,  $e * a = a$  ( $a * e = a$ ) for all  $a \in S$  if  $e * a = a * e = a$ , it is simply called neutral element  
 Usually denoted 0 for addition and 1 for multiplication

D5.4 If  $\langle S; * \rangle$  has left and right neutral element, they are equal

D5.5 Monoid: algebra  $\langle M; *; e \rangle$ , where  $*$  ass.,  $e$  neut. el.

D5.6 A left (right) inverse (element) of an element  $a$ , is an element  $b$ , such that  $b * a = e$  ( $a * b = e$ ), if  $b * a = a * b = e$ , simply inverse of  $a$

D5.7 If monoid  $\langle M; *; e \rangle$  has left and right inverse, they are equal

D5.7 Group: A group is an algebra  $\langle G; *; ^{-1}; e \rangle$  if it satisfies the following axioms: G1  $*$  is associative, G2  $e$  is a neutral element, G3 Every  $a \in G$  has an inverse element  $a^{-1}$

D5.8 A group (or monoid) is commutative or abelian if  $a * b = b * a$  for all  $a, b \in G$

D5.9 For a group we have: (i)  $\widehat{a^{-1}} = a$ , (ii)  $\widehat{a * b} = \widehat{b} * \widehat{a}$ , (iii) Left cancellation law:  $a * b = a * c \Rightarrow b = c$ , (iv) Right cancellation law:  $b * a = c * a \Rightarrow b = c$ , (v) The equation  $a * x = b$  has a unique solution  $x$  for any  $a, b$ . So does  $x * a = b$

D5.9 direct product of  $n$  groups  $\langle G_1; *_1 \rangle, \dots, \langle G_n; *_n \rangle$  is the algebra  $\langle G_1 \times \dots \times G_n; * \rangle$ , where the operation  $*$  is component-wise:  
 $(a_1, \dots, a_n) * (b_1, \dots, b_n) = (a_1 *_1 b_1, \dots, a_n *_n b_n)$

D5.4  $\langle G_1 \times \dots \times G_n; * \rangle$  is a group, where the neutral element and inversion operation are component-wise

D5.10 For two groups  $\langle G; *; ^{-1}; e \rangle$  and  $\langle H; \cdot; ^{-1}; e' \rangle$  a function  $\psi: G \rightarrow H$  is called group homomorphism if for all  $a, b \in G$   $\psi(a * b) = \psi(a) \cdot \psi(b)$   
 If  $\psi$  is a bijection, it's called isomorphism,  $G$  and  $H$  are isomorphic  $G \cong H$

D5.11 (i)  $\psi(e) = e'$ , (ii)  $\psi(a^{-1}) = \psi(a)^{-1}$  for all  $a$

D5.11  $H \subseteq G$  of a group  $\langle G; *; ^{-1}; e \rangle$  is called subgroup of  $G$  if  $\langle H; *; ^{-1}; e \rangle$  is a group, i.e.  $H$  is closed with respect to all operations (1)  $a * b \in H$  for all  $a, b \in H$ , (2)  $e \in H$ , (3)  $a^{-1} \in H$  for all  $a \in H$

There are always the trivial subgroups  $\{e\}$  and  $G$

D5.12 order of  $a \in G$  denoted  $\text{ord}(a)$  is the least  $m \geq 1$  such that  $a^m = e$  if  $m$  exists, otherwise  $\text{ord}(a) = \infty$

$\text{ord}(e) = 1$ , if  $\text{ord}(a) = 2$ ,  $a^{-1} = a$ ,  $a$  is called self-inverse

D5.6 In a finite group, every element has finite order

D5.13 For a finite group  $G$ ,  $|G|$  is called order of  $G$  if  $a \in G$  has finite order:  $a^m = a$   $\Leftrightarrow m \equiv \text{ord}(a) \pmod{m}$ ,  $m \in \mathbb{Z}$

D5.14 group generated by  $a$ :  $\langle a \rangle \stackrel{\text{def}}{=} \{a^n \mid n \in \mathbb{Z}\}$  for finite groups:  $\langle a \rangle \stackrel{\text{def}}{=} \{e, a, a^2, \dots, a^{\text{ord}(a)-1}\}$

D5.15  $G = \langle g \rangle$ ,  $G$  is a cyclic group,  $g$  is generator of  $G$  being cyclic is a special property of a group, can have many generators, if  $g$  is generator,  $g^{-1}$  is as well

D5.17 A cyclic group of order  $n$  is isomorphic to  $\langle \mathbb{Z}_n; \oplus \rangle$  and hence abelian

$\langle \mathbb{Z}_n; \oplus \rangle$ : standard notation of cyclic group of order  $n$

D5.8 (Lagrange): Let  $G$  be a finite group,  $H$  a subgroup of  $G$ :  $|H|$  divides  $|G|$

D5.9 For finite group  $G$ :  $\text{ord}(a)$  divides  $|G|$ , for all  $a \in G$

D5.10 For finite group  $G$ :  $a^{|G|} = e$ , for all  $a \in G$

D5.11 Every group of prime order is cyclic, every element except  $e$  is a generator

D5.16  $\mathbb{Z}_m^* \stackrel{\text{def}}{=} \{a \in \mathbb{Z}_m \mid \gcd(a, m) = 1\}$

D5.17 Euler function  $\varphi: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ ,  $\varphi(m) = |\mathbb{Z}_m^*|$  If  $p$  is a prime,  $\mathbb{Z}_p^* = \{1, \dots, p-1\} = \mathbb{Z}_p \setminus \{0\}$ ,  $\varphi(p) = p-1$

D5.12 if  $m = \prod_{i=1}^r p_i^{e_i}$  is the prime factorization of  $m$ ,  $\varphi(m) = \prod_{i=1}^r (p_i - 1) p_i^{e_i - 1}$

D5.13  $\langle \mathbb{Z}_m^*; \odot; ^{-1}; 1 \rangle$  is a group

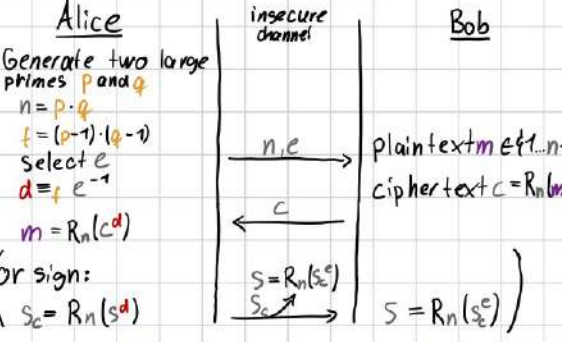
D5.14 (Fermat, Euler): For  $m \geq 2$ ,  $\gcd(a, m) = 1$ :  $a^{\varphi(m)} \equiv_m 1$ , for prime  $p$ :  $a^{p-1} \equiv_p 1$

D5.15  $\mathbb{Z}_m^*$  is cyclic iff  $m = 2, m = 4, m = p^e$  or  $m = 2p^e$ , where  $p$  is an odd prime and  $e \geq 1$

D5.16  $e$ -th root in a group,  $\gcd(e, |G|) = 1$ ,  $x \mapsto x^e$  is a bijection and the (unique)  $e$ -th root of  $y \in G$ ,  $y = x^e$ ,  $x = y^d$ , where  $d$  is the multiplicative inverse of  $e$  modulo  $|G|$  i.e.  $ed \equiv_{|G|} 1$

We can calculate  $d$ , given  $e$  with  $ed \equiv_{|G|} 1$  (by extended euclidian algorithm) iff we know  $|G|$

RSA Public-Key Encryption:



orange marked: just used for key-generation,  $\phi = \varphi(n) = \text{ord of } \mathbb{Z}_n^*$ , calculated via euler function,  $d$ : secret key, calculated by extended euclidean algorithm,  $(n, e)$ : public key

## 5 Rings and Fields

D5.18 Ring  $\langle R; +; \cdot; 0; 1 \rangle$  is an algebra with:

(i)  $\langle R; +; 0 \rangle$  is a commutative group, (ii)  $\langle R; \cdot; 1 \rangle$  is a monoid, (iii)  $a(b+c) = (ab) + (ac)$  and  $(b+c)a = (ba) + (ca)$  (left and right distr. law), A ring is called commutative if multiplication is commutative ( $ab = ba$ )

$\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$  are commutative rings

D5.17 For any ring: (i)  $0a = a0 = 0$ , (ii)  $(-a)b = -(ab)$ , (iii)  $(-a)(-b) = ab$ , (iv) If  $R$  is non-trivial (has more than one element)  $1 \neq 0$

D5.19 characteristic of a ring is order of 1 if finite, otherwise 0

DS20  $u \in R$  is called unit if  $u$  is (multiplicative) invertible, set of units of  $R$ :  $R^*$

LS18  $R^*$  is a group (multiplicative group of units)

In the following,  $R$  denotes a comm. ring

DS21  $a$  divides  $b$ , den.  $a|b$  if there exists  $a, c$ ,  $b = ac$ ,  $a$  a divisor of  $b$ ,  $b$  multiple of  $a$

LS19 In a comm. ring: (i) relation  $|$  is transitive, (ii) if  $a|b$  then  $a|bc$ , (iii) if  $a|b$  and  $a|c$ , then  $a|(b+c)$

DS22 greatest common divisor  $d$ :  $d|a \wedge d|b \wedge \forall c (c|a \wedge c|b) \rightarrow c|d$

DS23 zero divisor  $a$ :  $ab=0$ ,  $a \neq 0$ ,  $b \neq 0$

DS24 integral domain  $D$ : (non-trivial) comm. ring with no zero divisors,  $ab=0 \Rightarrow a=0 \vee b=0$

Ex:  $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$  are integral domains,  $\mathbb{Z}_m (m \neq p)$  not

LS20 if  $a|b$ ,  $b = ac$ ,  $c$  is unique den.  $c = \frac{b}{a}$  or  $c = b/a$  and called quotient

DS25 polynomial  $a(x)$  over comm. ring  $R$ :  $a(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i$

degree of  $a(x)$ , den.  $\deg(a(x))$  is greatest;  $\deg(0) = -\infty$ ,  $R[x]$  denotes set of polynomials (in  $x$ ) over  $R$

TS21 For any comm. ring  $R$ ,  $R[x]$  is a comm. ring

LS22 For integral domain  $D$ : (i)  $D[x]$  is integral domain, (ii) the deg. of the product of two polynomial is sum of deg., (iii)  $D[x]^* = D^*$

DS26 field (Körper): comm. ring  $F$ , with every non-zero element is unit:  $F^* = F \setminus \{0\}$

ring  $F$  is field iff  $\langle F \setminus \{0\}; \cdot, ^{-1}, 1 \rangle$  is abelian group

Ex:  $\mathbb{Q}, \mathbb{R}, \mathbb{C}$  are fields,  $\mathbb{Z}$  and  $R[x]$  (for any  $R$ ) not

TS23  $\mathbb{Z}_p$  is field iff  $p$  is prime

We denote field with  $p$  elements  $GF(p)$  rather than  $\mathbb{Z}_p$ , "GF" for Galois field

In field one can divide by any non-zero element

Ex. define a field:  $F = \{0, 1, A, B\}$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | A | B |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 |
| A | 0 | 1 | 0 |
| B | 0 | 0 | 1 |

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | A | B |
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 |
| A | 0 | 1 | 0 |
| B | 0 | 0 | 1 |

TS24 A field is an integral domain

DS27 A polynomial  $a(x) \in F[x]$  is monic if leading coefficient is 1

DS28 polynomial  $a(x)$  is irreducible if divisible only by constant polynomial and constant multiples of  $a(x)$  (Ex.  $x^2+1$  over  $\mathbb{R}$ , not  $\mathbb{C}$ )

Check irreducibility of  $a(x)$ : try all  $a(x)$  with  $\deg \leq \frac{d}{2}$

DS29 the greatest common divisor  $\gcd(a(x), b(x))$ : monic  $g(x)$  with largest deg. such that  $g(x)|a(x)$   $g(x)|b(x)$

TS25 For any  $a(x), b(x) \in F[x]$ ,  $b \neq 0$ , exist a unique  $q(x)$  (quotient) and unique  $r(x)$  (remainder),  $a(x) = b(x) \cdot q(x) + r(x)$  and  $\deg(r(x)) < \deg(b(x))$

remainder  $r(x)$ , can be denoted  $R_{b(x)}(a(x))$

Polynomial evaluation:  $R[x]$  can be interpreted as function  $R \rightarrow R$ :  $\alpha \mapsto a(\alpha)$

LS28 Polynomial evaluation is compatible with ring operations:  $C(x) = A(x) \pm B(x)$ , then  $c(\alpha) = b(\alpha) \pm b(\alpha)$

DS33 Roots (Wurzel/Nullstelle):  $\alpha \in R$  with  $a(\alpha) = 0$  is a root of  $a(x)$

LS29 for a field  $\alpha \in F$  is root of  $a(x)$  iff  $x - \alpha$  divides  $a(x)$

LS30 polynomial  $a(x)$  of deg 2 or 3 is irreducible iff it has no roots

TS31 a non-zero polynomial  $a(x) \in F[x]$  of degree  $d$  has at most  $d$  roots

LS32 Interpolation: a polynomial  $a(x) \in F[x]$  of deg most  $d$  is uniquely determined by  $d+1$  values of  $a(x)$ :  $a(\alpha_1), \dots, a(\alpha_d)$ , for distinct  $\alpha_i$ :  $a(x) \equiv_m b(x) \stackrel{\text{def}}{=} m(x) | (a(x) - b(x))$

LS33 Congruence modulo  $m(x)$  is an equivalence relation on  $F[x]$ , each class has a unique representative of deg less than  $\deg(m(x))$

DS34  $m(x)$  is polynomial of deg  $d$  over  $F$ ,  $F[x]_{m(x)} \stackrel{\text{def}}{=} \{a(x) \in F[x] | \deg(a(x)) < d\}$

LS34  $F$  has  $q$  elements,  $m(x)$  deg  $d$ ,  $|F[x]_{m(x)}| = q^d$

LS35  $F[x]_{m(x)}$  is a ring with addition and multiplication modulo  $m(x)$

LS36 congruence relation  $a(x) b(x) \equiv_m 1$

for given  $a(x)$  has solution  $b(x) \in F[x]_{m(x)}$  iff  $\gcd(a(x), m(x)) = 1$ , the solution is unique  $F[x]_{m(x)}^* = \{a(x) \in F[x]_{m(x)} | \gcd(a(x), m(x)) = 1\}$

TS37  $F[x]_{m(x)}$  is a field iff  $m(x)$  is irreducible

## ⑤ Error-Correcting Codes

DS35  $(n, k)$ -encoding function  $E$ , for alphabet  $A$  is injective function that maps  $(a_0, \dots, a_{k-1}) \in A^k$  of  $k$  (information) symbols to  $(c_0, \dots, c_{n-1}) \in A^n$  of  $n > k$  (encoded) symbols in  $A$  called codeword  $E: A^k \rightarrow A^n: (a_0, \dots, a_{k-1}) \mapsto E((a_0, \dots, a_{k-1})) = (c_0, \dots, c_{n-1})$   $C = \text{Im}(E) = \{E((a_0, \dots, a_{k-1})) | a_0, \dots, a_{k-1} \in A\}$

DS36  $(n, k)$ -error correcting code over  $A$  with  $|A| = q$  is subset of  $A^n$  of cardinality  $q^k$  It's natural to use  $A = \{0, 1\}$ , but often for reasons like efficiency one considers larger units ex. bytes  $A = \{0, \dots, 255\}$

DS37 Hamming distance between two strings of equal length over finite  $A$  is number of positions, where the two strings differ

DS38 minimum distance  $d_{\min}(C)$ , is the minimal Hamming distance between any two codewords

DS39 decoding function  $D: A^n \rightarrow A^k$  a good decoding function takes any  $(r_0, \dots, r_{n-1}) \in A^n$  and decodes it to most plausible  $(a_0, \dots, a_{k-1})$ , efficiently

DS40 a decoding function  $D$  is  $t$ -error correcting if  $D((r_0, \dots, r_{n-1})) = (a_0, \dots, a_{k-1})$  if  $(r_0, \dots, r_{n-1})$  has Hamming distance at most  $t$  from  $E((a_0, \dots, a_{k-1}))$   $C$  is  $t$ -error correcting if there exists  $t$ -error correcting  $D$

TS41 code  $C$  with min dist  $d$  is  $t$ -error correcting iff  $d \geq 2t + 1$  (".")

DS42 Let  $A = GF(q)$ ,  $\alpha_0, \dots, \alpha_{k-1}$  arbitrary distinct elements of  $GF(q)$ , encoding function:  $E((a_0, \dots, a_{k-1})) = (a(\alpha_0), \dots, a(\alpha_{k-1}))$ , where  $a(x)$  is polynomial  $a(x) = a_{k-1}x^{k-1} + \dots + a_1x + a_0$ , the min dist is  $n - k + 1$

Polynomial codes as described are used in CD's

## ⑥ Logic

All objects of study are described in well-defined syntax, syntactic objects are finite strings over a alphabet  $\Sigma$ , denoted  $\Sigma^*$

In this section two types of mathematical objects: mathematical statements, proofs

$S \subseteq \Sigma^*$ : set of (syntactic representations of) mathematical statements

$\mathcal{P} \subseteq \Sigma^*$ : set of (syntactic representations of) mathematical proof strings

Every  $s \in S$  is either true or false the truth function  $\mathcal{T}: S \rightarrow \{0, 1\}$  assigns each  $s$  its truth value  $\mathcal{T}(s)$ ,  $\mathcal{T}$  defines the semantics

Every  $p \in \mathcal{P}$  is either (valid) proof for  $s \in S$  or not verification function  $\Phi: S \times \mathcal{P} \rightarrow \{0, 1\}$ ,  $\Phi(s, p) = 1$  means  $p$  is valid proof for  $s$

Without loss of generality:  $S = \mathcal{P} = \{0, 1\}^*$

DS1 proof system is a quadruple  $\Pi = (S, \mathcal{P}, \mathcal{T}, \Phi)$

DS2 proof system  $\Pi$  is sound (korrekt) if no false statement has a proof ( $\Phi(s, p) = 1 \Rightarrow \mathcal{T}(s) = 1$ )

DS3 proof system  $\Pi$  is complete (vollständig) if every true statement has a proof ( $\mathcal{T}(s) = 1 \Rightarrow \exists p (\Phi(s, p) = 1)$ )  $\Phi$  needs to be efficiently computable

DS4 syntax of a logic: defines alphabet  $\Lambda$  (of allowed symbols) and specifies which strings in  $\Lambda^*$  are formulas

DS5 semantics of logic: defines a function free which assigns to each formula  $F = (t_1, t_2, \dots, t_k) \in \Lambda^*$  a subset  $\text{free}(F) \subseteq \{1, \dots, k\}$  of indices, if  $i \in \text{free}(F)$   $t_i$  occurs free in  $F$

DS6 interpretation: consists of:  $\mathcal{Z} \subseteq \Lambda$ , domain (set of possible values) and function that assigns each  $z \in \mathcal{Z}$  a value of its domain

Domains are often defined in terms of a universe  $U$

DS7 An interpretation is suitable if it assigns a value to all  $\beta \in \Lambda$  occurring free in  $F$

DS8 semantics of a logic also defines a function  $\sigma$ ,  $\sigma(F, \mathcal{A})$  is a truth value in  $\{0, 1\}$ , one often writes  $\mathcal{A}(F)$  instead of  $\sigma(F, \mathcal{A})$ , "truth val of  $F$  under interp.  $\mathcal{A}$ "

D6.9 A (suitable) interpretation  $A$  for which  $F$  is true is called **model** for  $F$ :  $A \models F$ , more general for set of formulas  $M$ :  $A \models M$ , if  $A$  is not model for  $M$ :  $A \not\models M$

D6.10  $F$  (or  $M$ ) is **satisfiable** if there exists a model for it, **unsatisfiable** otherwise,  $\perp$  (not a formula) used for unsat. formula

D6.11  $F$  is called **tautology** (=valid) if true for every suitable interpretation, den.  $\top$ .  $\perp$  is called **falsum**,  $\top$  is called **verum**

D6.12  $G$  is **logical consequence** of  $F$  (or  $M$ ), den.  $F \models G$  (or  $M \models G$ ), every interpretation suitable for both, which is model for  $F$  (or  $M$ ) is also model for  $G$

D6.13  $F$  and  $G$  are **equivalent**:  $F \models G \text{ and } G \models F$

D6.14 if  $F$  is tautology one writes  $\models F$ .  $F$  is **unsatisfiable**  $F \models \perp$

D6.15 If  $F, G$  are formulas,  $\neg F, (F \wedge G), (F \vee G)$  are also  
 $\wedge$ : conjunction,  $\vee$ : disjunction,  $F \rightarrow G$  stands for  $\neg F \vee G$ ,  $F \leftrightarrow G$  stands for  $(F \wedge G) \vee (\neg F \wedge \neg G)$

D6.16  $A((F \wedge G)) = 1$  iff  $A(F) = 1$  and  $A(G) = 1$   
 $A((F \vee G)) = 1$  iff  $A(F) = 1$  or  $A(G) = 1$   
 $A(\neg F) = 1$  iff  $A(F) = 0$

L6.1 For any formulas  $F, G, H$  we have:

- 1)  $F \wedge F \equiv F$  and  $F \vee F \equiv F$  (idempotence)
- 2)  $F \wedge G \equiv G \wedge F$  and  $F \vee G \equiv G \vee F$  (commutativity)
- 3)  $(F \wedge G) \wedge H \equiv F \wedge (G \wedge H)$  and  $(F \vee G) \vee H \equiv F \vee (G \vee H)$  (associativity)
- 4)  $F \wedge (F \vee G) \equiv F$  and  $F \vee (F \wedge G) \equiv F$  (absorption)
- 5)  $F \wedge (G \vee H) \equiv (F \wedge G) \vee (F \wedge H)$  (distributive law)
- 6)  $F \vee (G \wedge H) \equiv (F \vee G) \wedge (F \vee H)$  (distributive law)
- 7)  $\neg \neg F \equiv F$  (double negation)
- 8)  $\neg (F \wedge G) \equiv \neg F \vee \neg G$  and  $\neg (F \vee G) \equiv \neg F \wedge \neg G$  (de Morgan's rule)
- 9)  $F \vee \top \equiv \top$  and  $F \wedge \top \equiv F$  (tautology rules)
- 10)  $F \vee \perp \equiv F$  and  $F \wedge \perp \equiv \perp$  (unsatisfiability rules)
- 11)  $F \vee \neg F \equiv \top$  and  $F \wedge \neg F \equiv \perp$

L6.2  $F$  is tautology iff  $\neg F$  is unsatisfiable

L6.3 The following three statements are equivalent:  
1.  $\{F_1, \dots, F_k\} \models G$ , 2.  $(F_1 \wedge \dots \wedge F_k) \rightarrow G$  is a tautology,  
3.  $\{F_1, \dots, F_k, \neg G\}$  is unsatisfiable

## ⑥ Logical Calculi (Kalküle)

A proof of a Theorem (axioms  $\models T$ ) should be purely syntactic, syntactic verification can be done by computer (without reasoning between lines) verify proof: dumb process, devising proof: intelligent, There is a trade-off between simplicity (small set of rules) and versatility (large set of rules)

Hilbert-Style calculi (most intuitive, discussed here):

D6.17 **derivation rule** (=inference rule, Schlussregel): rule for deriving a formula from a set of formulas (precondition or premises) by rule  $R$ .  $\{F_1, \dots, F_k\} \vdash_R G$ , Formally  $R$  is relation from power set of the set of formulas to the set of formulas

$\{F_1, \dots, F_k\} \vdash_R G$  sometimes written as  $\frac{F_1 \dots F_k}{G} (R)$

D6.18 Application of derivation rule  $R$  to set  $M$  of formulas  
1. select  $N \subseteq M$  such that  $N \vdash_R G$   
2. Add  $G$  to  $M$  i.e. replace  $M$  by  $M \cup \{G\}$

D6.19 (logical) calculus  $K$  is a finite set of derivation rules:  $K = \{R_1, \dots, R_n\}$

D6.20 A **derivation** of  $G$  from  $M$  in  $K$  is finite sequence of applications of rules in  $K$  leading to  $G$ :  $M_0 := M$ ,  $M_i := M_{i-1} \cup \{G_i\}$ , where  $N \vdash_{R_j} G_i$  for  $N \subseteq M_{i-1}$ ,  $R_j \in K$ ,  $G_n = G$ , we write  $M \vdash_K G$  if there is a der. of  $G$  from  $M$  in  $K$

D6.21  $R$  is **correct** if  $M \vdash_R F \Rightarrow M \models F$

D6.22  $K$  is **sound** (widerspruchsfrei) or **correct** if  $M \vdash_K F \Rightarrow M \models F$ ,  $K$  is **complete** if  $M \models F \Rightarrow M \vdash_K F$

One writes  $\vdash_K F$  if  $F$  can be derived from empty set, if  $\vdash_K F$  for sound calculus

$F$  is tautology

derivation rules for propositional logic and logics containing propositional logic: -basic equivalences of L6.1,  $F \wedge G \vdash F$ ,  $F \wedge G \vdash G$ ,  $\{F, G\} \vdash F \wedge G$ ,  $F \vdash F \vee G$ ,  $F \vdash G \vee F$ ,  $\{F, F \rightarrow G\} \vdash G$ ,  $\{F \vee G, F \rightarrow H, G \rightarrow H\} \vdash H$ ,  $\vdash F \vee \neg F$  (to start from empty set),  $\vdash \neg(F \leftrightarrow \neg F)$ ,  $\{F \vee G, \neg G\} \vdash F$  (proof of contradiction)

In sound calculus:  $F \vdash_K G \Rightarrow \models (F \rightarrow G)$ , one could extend the calculus by  $\vdash (F \rightarrow G)$ , more general  $\{F_1, F_2\} \vdash_K G \Rightarrow \models ((F_1 \wedge F_2) \rightarrow G)$

## ⑥ Propositional Logic

D6.23 Syntax) Atomic formula symbol from  $A_i, i \in \mathbb{N}$   
- An atomic formula is a formula, - D6.15

In propositional logic, all atomic formulas are the free symbols

D6.24 (Semantics) set  $Z$  of atomic formulas, interpretation  $A$  called **truth assignment**  $A: Z \rightarrow \{0, 1\}$ ,  $A$  is suitable for  $F$  if  $Z$  contains all atomic formulas in  $F$ , + D6.16

D6.25 **literal**: atomic formula or it's negation

D6.26  $F$  is in **conjunctive normal form (CNF)** if  $F = (L_{11} \vee \dots \vee L_{1n_1}) \wedge \dots \wedge (L_{n_1} \vee \dots \vee L_{n_m})$

D6.27  $F$  is in **disjunctive normal form (DNF)** if  $F = (L_{11} \wedge \dots \wedge L_{1n_1}) \vee \dots \vee (L_{n_1} \wedge \dots \wedge L_{n_m})$

L6.4 Every  $F$  is equivalent to a formula in CNF/DNF (using the truth table, DNF: where 1, CNF: neg of where 0)

**Resolution Calculus**: important logical calculus, consists of exactly one derivation rule, purpose: prove unsatisfiability, assumes formula is in CNF, works with CNF equivalent objects, namely sets of clauses

D6.28 **clause** set of literals

D6.29 set of clauses:  $\mathcal{K} \stackrel{\text{def}}{=} \{\{L_{11}, \dots, L_{1n_1}\}, \dots, \{L_{n_1}, \dots, L_{n_m}\}\}$   
Set of clauses for  $M = \{F_1, \dots, F_k\}: \mathcal{K}(M) \stackrel{\text{def}}{=} \bigcup_{i=1}^k \mathcal{K}(F_i)$   
clause stands for disjunction (OR), set of clauses stands for conjunction (AND)

**empty clause**: unsatisfiable, empty set of clauses:  $\top$

D6.30 clause  $K$  is **resolvent** of clauses  $K_1$  and  $K_2$

if there is a  $L \in K_1$ ,  $\neg L \in K_2$  and  $K = (K_1 \setminus \{L\}) \cup (K_2 \setminus \{\neg L\})$   
important: resolution steps one by one

**resolution step**: take  $K_1, K_2 \in \mathcal{K}$ , compute resolvent  $K$ , add  $K$  to  $\mathcal{K}$ , Example of a informal but correct representation:  
 $\mathcal{K}(F) = \{\{A, B\}, \{A, B\} \rightarrow A, C\} \rightarrow A, C\} \Rightarrow F$  is unsatisfiable

$\{K_1, K_2\} \vdash_{\text{res}} K$ ,  $\text{Res} = \{\text{res}\}$ ,  $\mathcal{K} \vdash_{\text{res}} K$  if  $K$  can be derived from  $\mathcal{K}$  in a finite number of steps

L6.5 The resolution calculus is sound

T6.6  $M$  is unsatisfiable iff  $\mathcal{K}(M) \vdash_{\text{res}} \emptyset$

## ⑥ Predicate Logic

D6.31 Syntax: variable symbol:  $x_i, i \in \mathbb{N}$ , function symbol:  $f^{(k)}, k \in \mathbb{N}$ ,  $k$  den. number of arguments,  $k=0$  are **constants**, predicate symbol:  $P^{(k)}, k \in \mathbb{N}$ , term: variables/funs, formula (inductively):  $P^{(k)}$  is atomic formula, D6.16,  $\forall x: F, \exists x: F$  are formulas

$\forall$ : universal quantifier ("all quantifier"),  $\exists$ : existential quantifier

D6.32 a variable  $x$  is **bound**, if occurring in (sub-)formula of  $\forall x G$  or  $\exists x G$ , free otherwise,  $F$  with no free vars: **closed**

D6.33  $F[x \mapsto t]$ :  $F$ , all free occurrences of  $x$  substituted by  $t$

D6.34 **interpretation** (=structure)  $A = (U, \Phi, \Psi, \xi)$ , where  $U$ : universe,  $\Phi$ : assigning to each function symbol a function  $(U^k \rightarrow U)$ ,  $\Psi$  assigning to each predicate symbol a function  $(U^k \rightarrow \{0, 1\})$ ,  $\xi$ : assigning to each variable symbol a value in  $U$

For notational convenience:  $\exists^A, P^A, x^A, U^A$

D6.35  $A$  is suitable if it defines all function symbols, predicate symbols and freely occurring variables

D6.36 **Semantics**: value for  $A(t)$  ( $n$  formula) is defined recursively: if  $t$  is a var  $A(t) = \xi(t)$ , if  $t$  is  $k$ -ary func  $A(t) = \Phi(t)(A(t_1), \dots, A(t_k))$ , truth-value of  $A(F)$  is defined recursively by D6.16, if  $F$  is  $k$ -ary predicate  $A(F) = \Psi(P)(A(t_1), \dots, A(t_k))$ ,  $A(\forall x G) = \begin{cases} 1 & \text{if } A(x \mapsto u)(G) = 1 \text{ for all } u \text{ in } U \\ 0 & \text{else} \end{cases}$ ,  $A(\exists x G) = \begin{cases} 1 & \text{if } A(x \mapsto u)(G) = 1 \text{ for some } u \text{ in } U \\ 0 & \text{else} \end{cases}$

Note: syntax does not require nested quantified vars to be distinct, but we will avoid such overload

